Short Distance Flights & High Speed Rail Alternatives in Europe

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Motivation

- **Course**: Geospatial Data Science & Economic Spatial Models
- **Purpose**: Design a final capstone project for the course, implementing GIS concepts from the term combined with spatial models.
- **Research Question:** Which short-haul European flight routes are substitutable by existing high-speed rail, and what policy levers make that substitution welfare-improving?

Data & Methodology

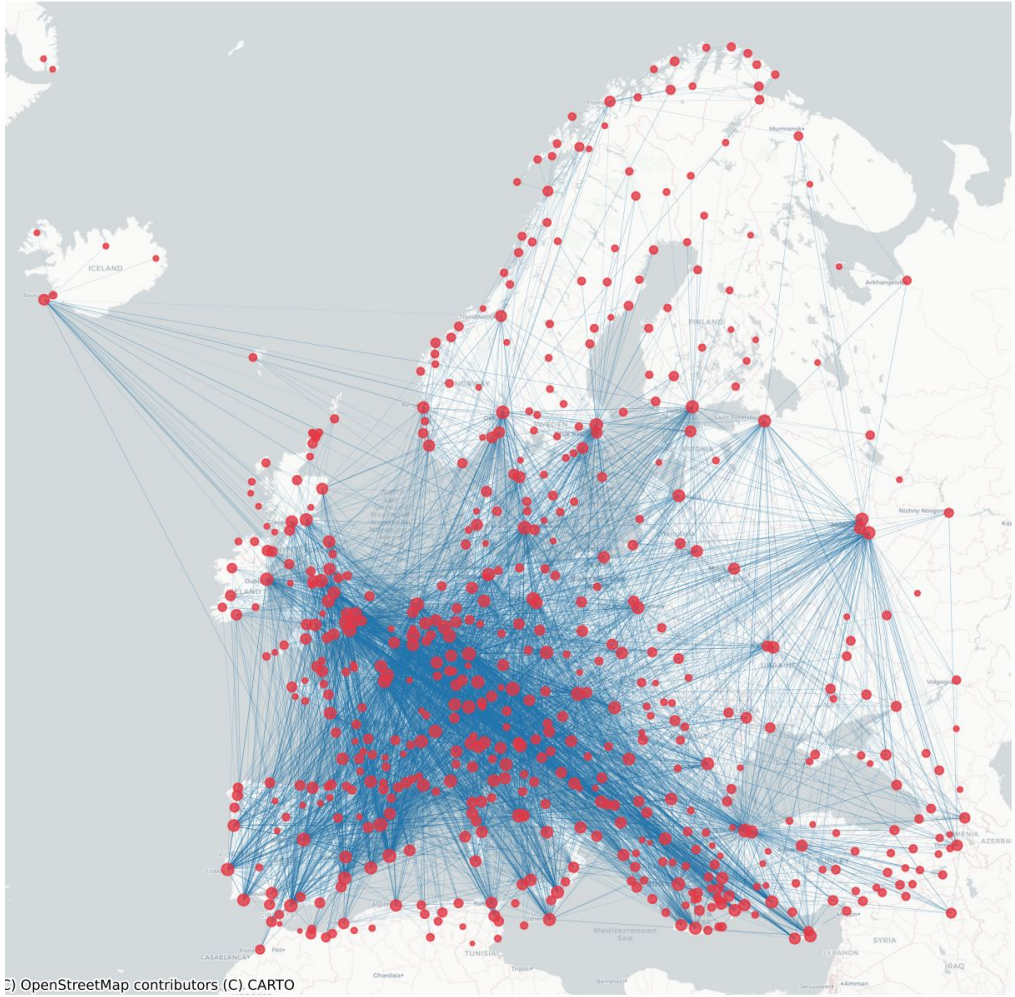
- **Data**
  - GL2017 EU Rail, OpenFlights (airports + routes), Eurostat avia_paoa (passenger volume)
- **Methods**
  - **Distance Calculation:** Used Haversine formula to determine flight distance between airport pairs.
  - **Categorization:** Grouped flights into distance buckets (e.g., <300km, 300-500km, >500km).
  - **Substitution Analysis:** Defined "replaceable" flights as those under 500km with existing or potential rail connectivity.
  - **Emission Estimation:** Applied CO2 emission factors per passenger kilometer based on aircraft type or industry averages.

European Rail Network: Conventional & High Speed

- First, located HSR data from GL2017 EU Rail dataset (outdated, but easiest to access for quick analysis)
- Tried OSM queries based on HSR flags or max speeds, but had timeout issues

Data: GL2017 EU Rail, OpenFlights, Eurostat avia_paoa

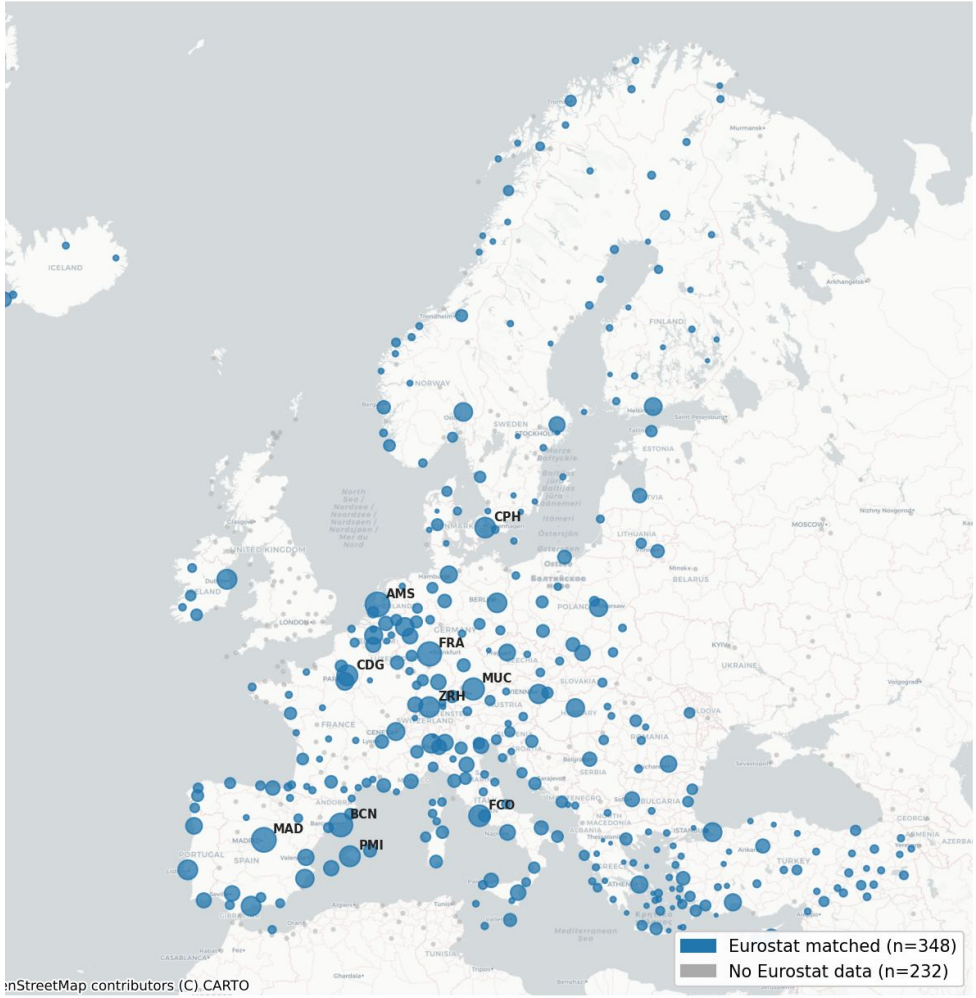
European Flight Network (16,758 routes | 580 airports)



- Next, utilized OpenFlight data to map airports & connecting flight routes. Messy!

Data: GL2017 EU Rail, OpenFlights, Eu

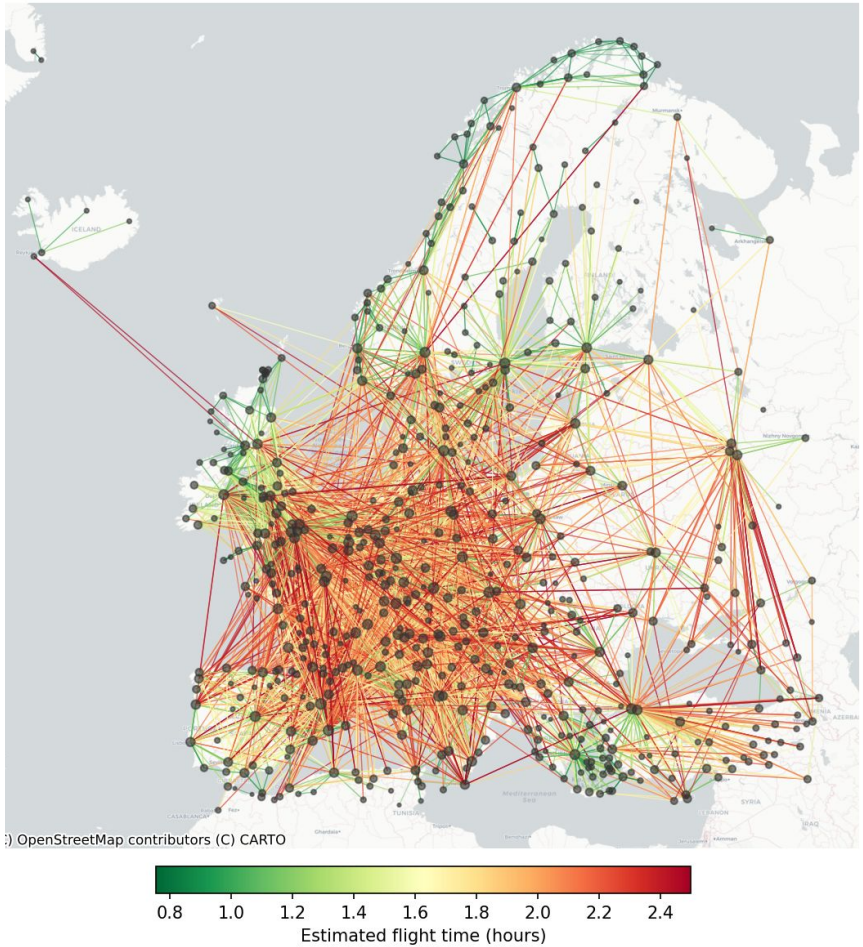
European Civil Airports — Annual Passenger Traffic (2025)



Data: GL2017 EU Rail, OpenFlights, Eurostat a

- Joined in Eurostat data (avia_paoa) for total passenger counts. Looked at the most recent quarterly estimates
- Used passenger data to filter out non-commercial airports; we are not interested in smaller military or private airports for our analysis

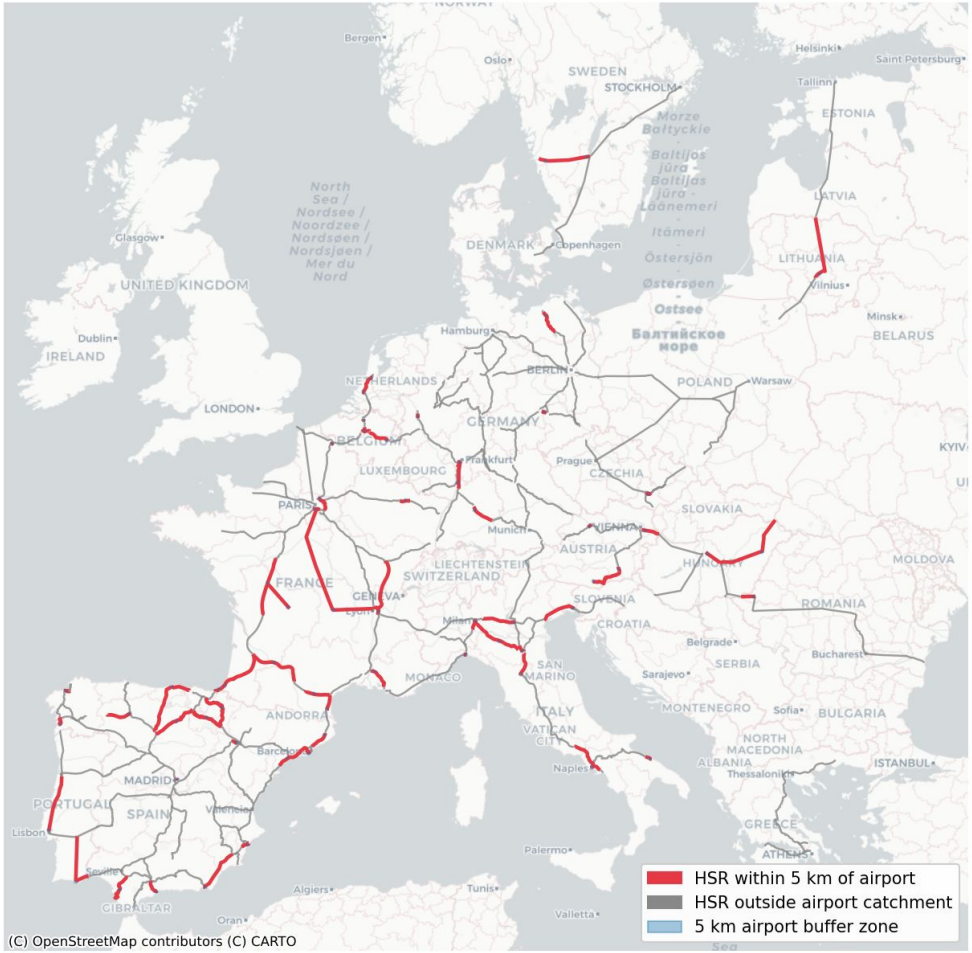
Short-Haul Flight Routes (≤ 2.5 h | n=10,694 routes)
Green = short hop | Red = approaching 2.5 h



- Used Haversine formula to calculate flight time. Literature suggests flight time is strong metric to identify flights best suited for HSR alternatives.
- Also messy, but some patterns emerging.

Data: GL2017 EU Rail, OpenFlights, Eurostat avia_pa

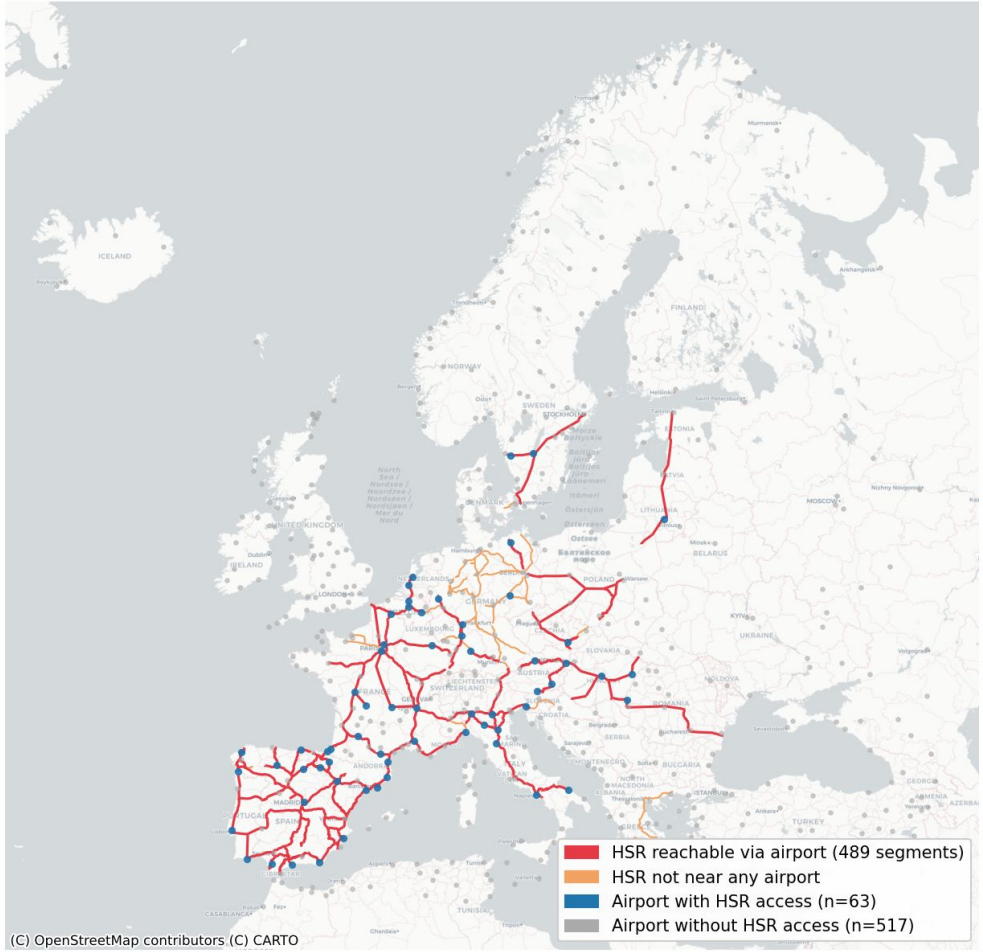
Methodology: 5 km Airport Buffer ∩ HSR Network
Identifies HSR segments directly accessible from airport terminals



- Applied 5m buffer around airports to detect any HSR edge (not just stations, potential limitation here)

Data: GL2017 EU Rail, OpenFlights, Eur

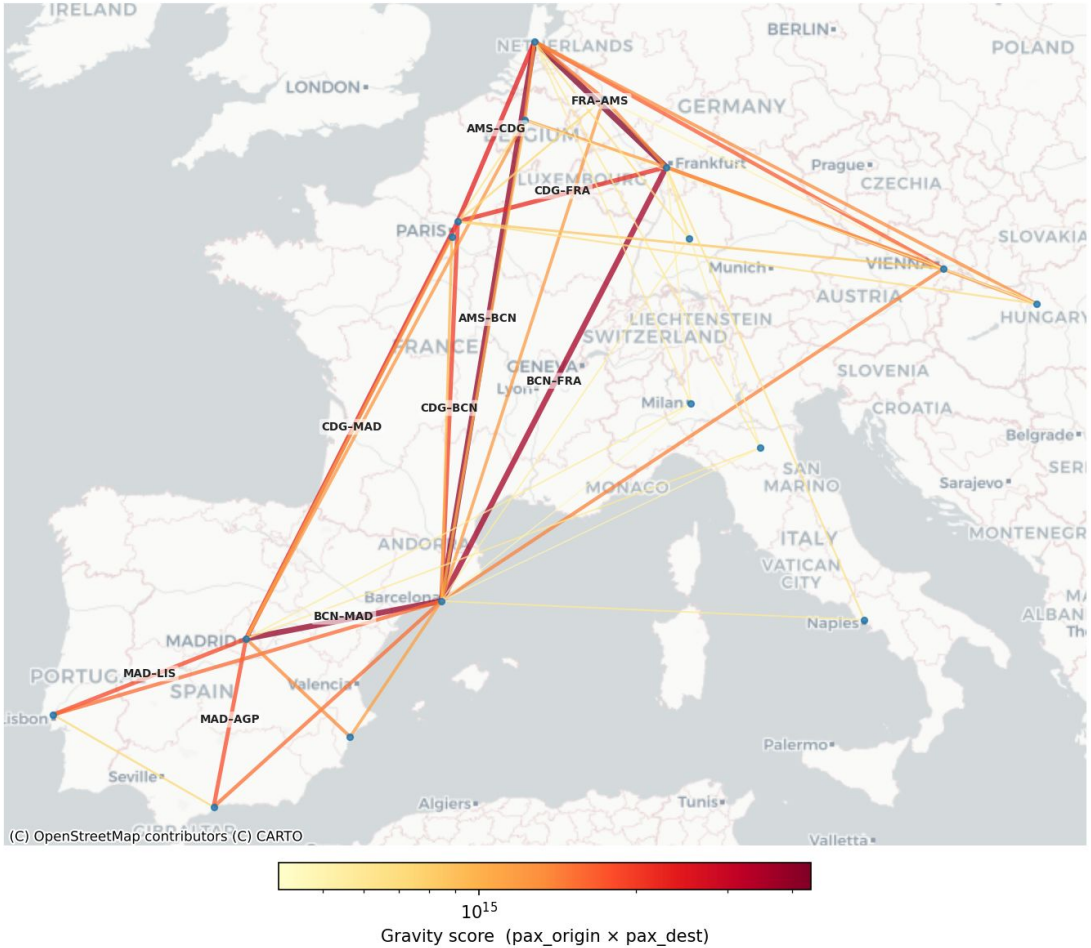
HSR Network Reachable from Civil Airports (flood-fill from 5 km catchment)
Airports trigger access to entire geometrically-connected HSR sub-network



Data: GL2017 EU Rail

- Next used flood-fill algorithm to move from a single segment to a functional network.
  - At each iteration, accessible segments were buffered by 200 meters to "jump" small gaps in the geometry (e.g., station breaks or data digitizing errors).
  - The process repeats until no new touching segments are found, identifying the entire reachable HSR sub-network.

Top 50 Short-Haul Routes Substitutable by HSR
Both airports ≤ 5 km from HSR | Flight ≤ 2.5 h | Ranked by passenger gravity



Preliminary finding: Most subsitutiable short haul routes

- Classification Criteria
  - Flight time is <2.5 hours
  - Both origin and destination airports are within 5 km of the HSR network.

Data: GL2017 EU R

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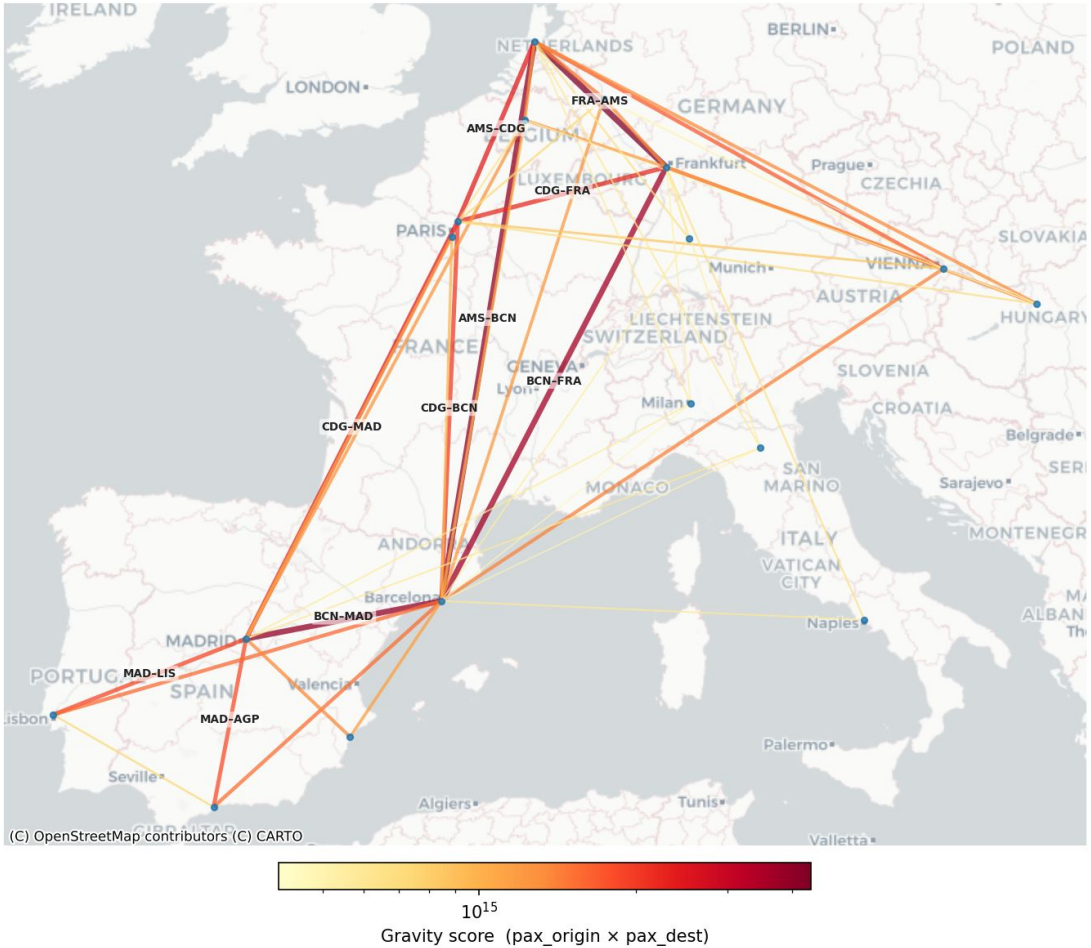
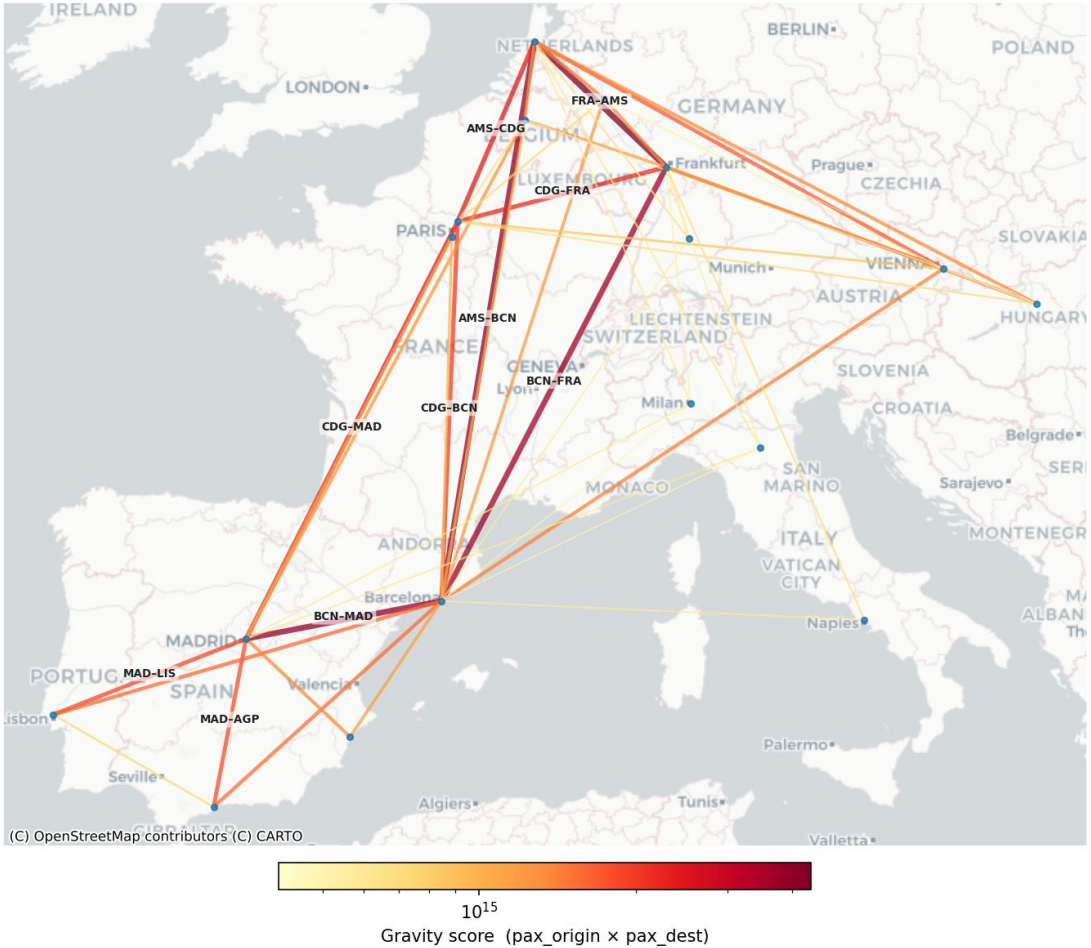


Preliminary finding: Most subsitutiable short haul routes

- Out of 10,694 total short-haul routes analyzed, only 187 (1.7%) currently meet the dual-HSR access criteria.
- This highlights a significant gap between geographic proximity and functional HSR connectivity.

Data: GL2017 EU R

Top 50 Short-Haul Routes Substitutable by HSR
Both airports ≤ 5 km from HSR | Flight ≤ 2.5 h | Ranked by passenger gravity



Gravity Score

- Proxy for route-level passenger counts
- Multiply annual passenger totals for origin by destination
- Follows standard bilateral transport models where the "pull" between two hubs is proportional to their respective sizes.

Data: GL2017 EU R

The Economic Model: Carbon-Adjusted Market Access

- Adapted from the Allen & Arkolakis (2014) trade model to measure "Clean Connectivity."
- **Market Access (MA_i)**: Quantifies how well an airport is connected to the rest of the network, weighted by destination "mass" (passengers) and the cost to get there.
- **The Bilateral Cost (L_{ij}):** "Generalized Cost" function that combines two factors:
  - Time Cost: Travel time multiplied by the Value of Time (€30/hr).
  - Carbon Cost: Total emissions multiplied by a Carbon Price (€100/t).
- **The Counterfactual Scenario:** What happens to connectivity if we swap high-emission aviation costs with low-emission HSR costs on all 187 substitutable routes?
  - Result: The change in Market Access (Delta MA) measures the potential gain in "clean" economic integration.

The Economic Model: Carbon-Adjusted Market Access

From Jupyter notebook:

Following Allen & Arkolakis (2014), each airport's market access is:

$$MA_i = \sum_{j \neq i} Y_j \cdot \tau_{ij}^{-\theta}$$

where Y_j is annual passengers (Eurostat), τ_{ij} is the carbon-adjusted bilateral cost, and θ is the travel elasticity. The bilateral cost embeds both time and carbon:

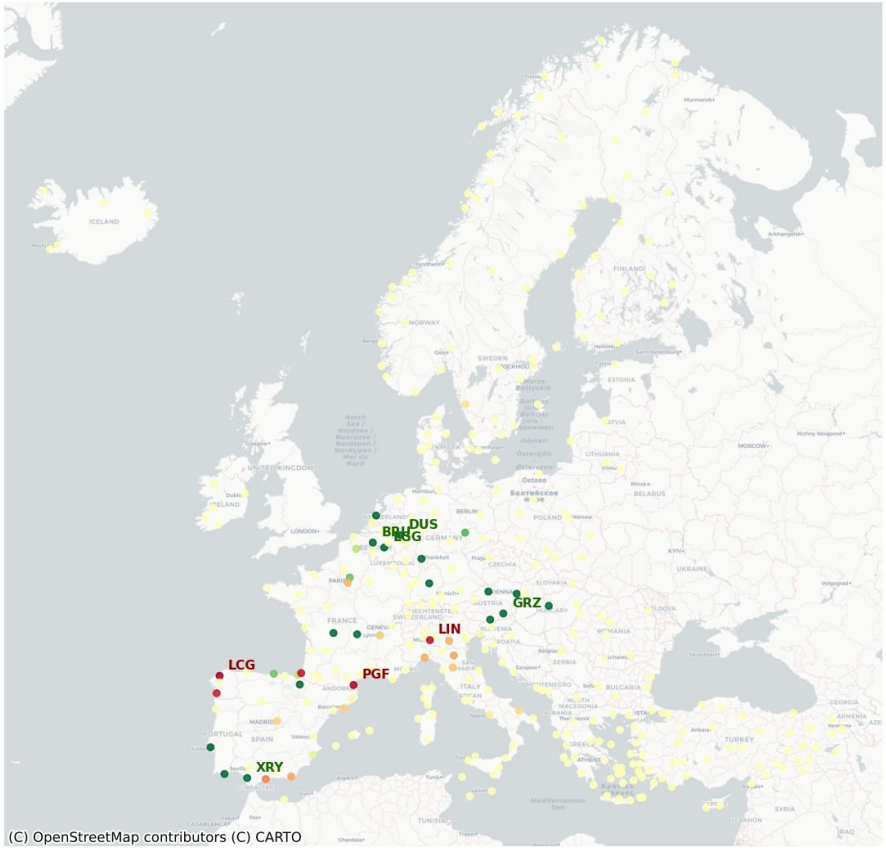
$$\tau_{ij} = t_{ij} + \lambda \cdot \mathrm{CO}_2^{ij} \quad \text{where} \quad \lambda = \frac{\text{carbon price}}{\text{value of time}} = \frac{€\ 100/\mathrm{t}}{€\ 30/\mathrm{hr}}$$

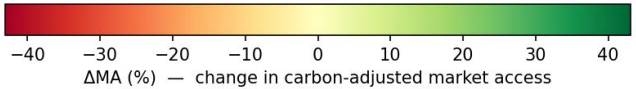
The counterfactual replaces aviation cost with HSR cost on all substitutable routes. ΔMA_i measures the change in clean connectivity.

The Economic Model: Parameters & Assumptions

- **Travel Elasticity (θ)**
  - Tested at values of 3, 5, and 8.
  - Insight: Results are directionally robust (connectivity always improves) but magnitude-sensitive (higher θ amplifies the benefits of switching to rail).
- **Emission Intensities**
  - Aviation: 255 g/pax-km (includes radiative forcing multipliers)
  - HSR: 14 g/pax-km (European average).
- **Valuation**
  - Carbon Price: Set at €100/tonne (aligned with future EU climate targets rather than current spot prices)
  - Time Value: Fixed at €30/hour across all passenger types.

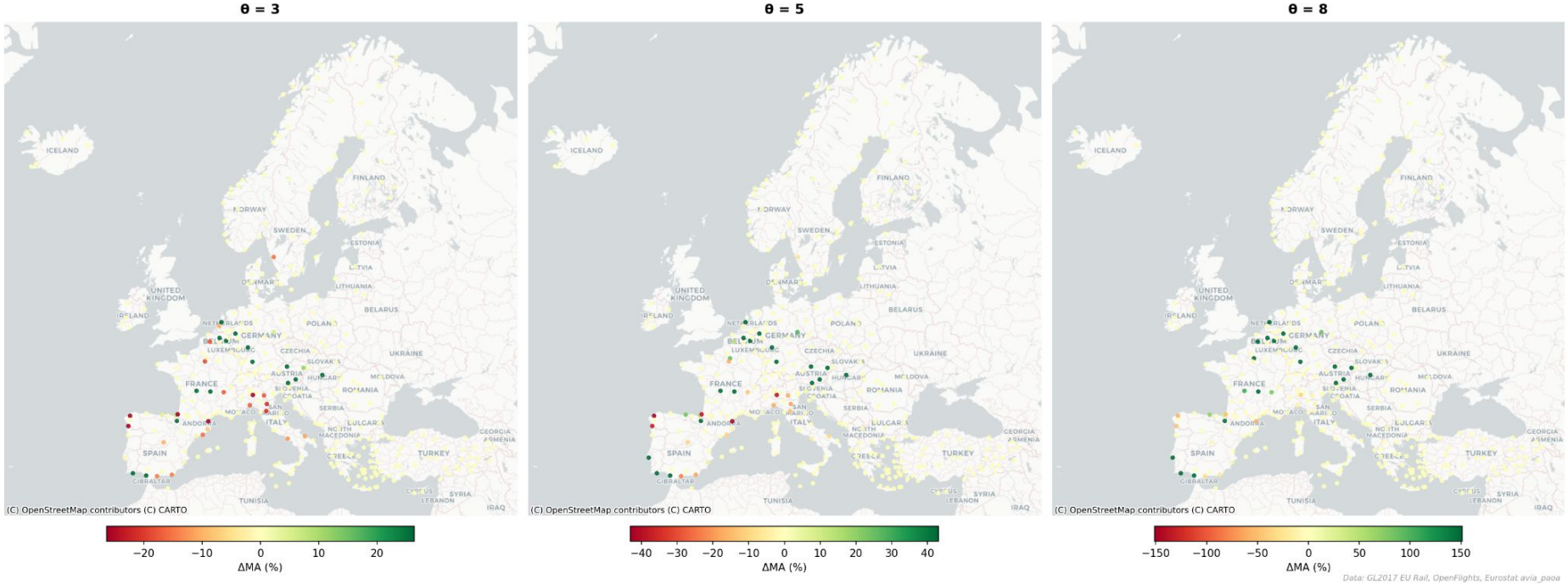
Change in Carbon-Adjusted Market Access (θ = 5)
Allen & Arkolakis (2014) | Counterfactual: HSR substitution on eligible routes





Data: GL2017 EU Ra

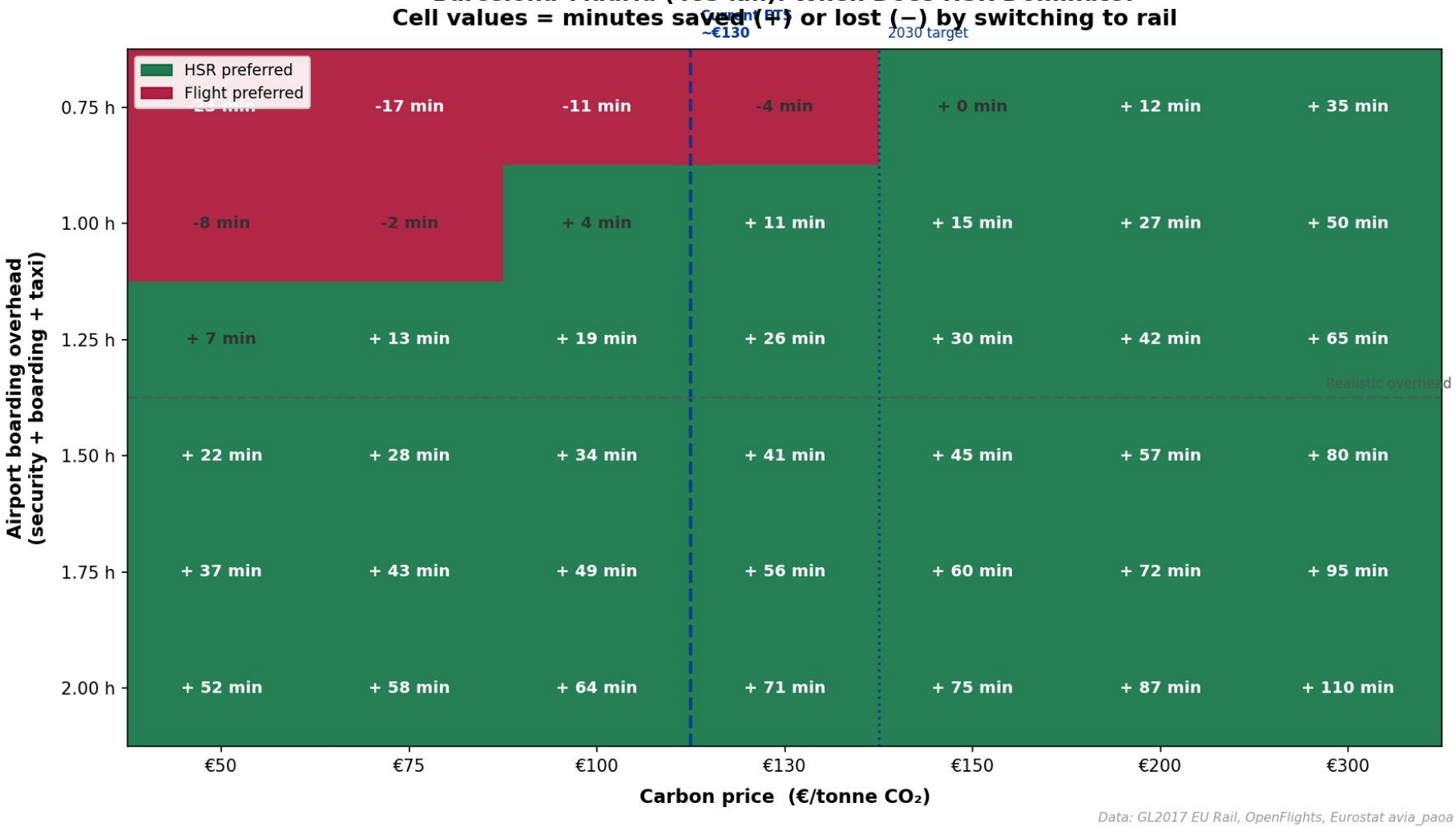
Sensitivity to Trade Elasticity θ — Change in Carbon-Adjusted Market Access



Data: GL2017 EU Rail, OpenFlights, Eurostat avia_paoa

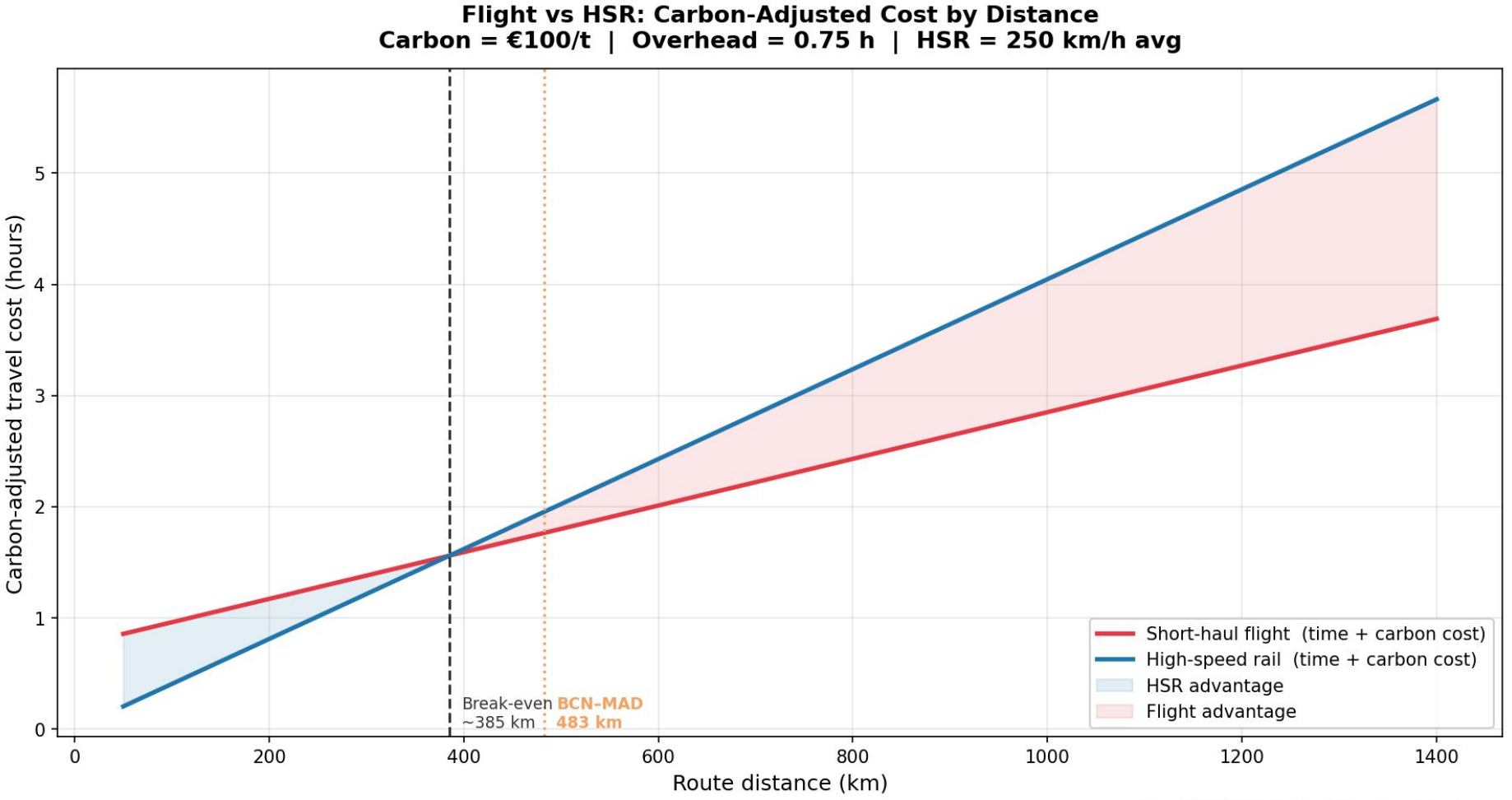
Barcelona–Madrid: Case Student & Policy Sensitivity

- **Why BCN–MAD if there is already HSR?**
  - Highly documented example of air-to-rail modal shift.
  - After AVE opened in 2008, air passengers on the route fell ~60% within 3 years (Eurostat).
  - Demonstrates that HSR can genuinely substitute for short-haul aviation at scale.
- **Route profile**
  - Distance: 483 km, above the model's baseline break-even (~385 km)
  - Pre-AVE dominated by Iberia/Vueling with ~6M annual air passengers. Post-AVE, Renfe holds over 75% combined modal share.
- **Why it sits near the tipping point**
  - At 483 km, BCN–MAD is too long for HSR to win on time alone under baseline assumptions.
  - Carbon pricing and airport friction are the two levers that determine whether HSR dominates, making it an ideal stress-test for the model's sensitivity.
- **What follows**
  - The heatmap maps HSR competitiveness across carbon price (€50–300/t) and airport overhead (0.75–2.0 h).
  - Cell values show minutes saved or lost by switching from flight to rail.


Barcelona–Madrid (483 km): When Does HSR Dominate?
Cell values = minutes saved (+) or lost (−) by switching to rail
Current ETS ~€130
2030 target
HSR preferred
Flight preferred
Airport boarding overhead (security + boarding + taxi)
0.75 h
1.00 h
1.25 h
1.50 h
1.75 h
2.00 h
-17 min
-11 min
-4 min
+ 0 min
+ 12 min
+ 35 min
-8 min
-2 min
+ 4 min
+ 11 min
+ 15 min
+ 27 min
+ 50 min
+ 7 min
+ 13 min
+ 19 min
+ 26 min
+ 30 min
+ 42 min
+ 65 min
Realistic overhead
+ 22 min
+ 28 min
+ 34 min
+ 41 min
+ 45 min
+ 57 min
+ 80 min
+ 37 min
+ 43 min
+ 49 min
+ 56 min
+ 60 min
+ 72 min
+ 95 min
+ 52 min
+ 58 min
+ 64 min
+ 71 min
+ 75 min
+ 87 min
+ 110 min
€50
€75
€100
€130
€150
€200
€300
Carbon price (€/tonne CO₂)
Data: GL2017 EU Rail, OpenFlights, Eurostat avia_paoa

Barcelona–Madrid: Policy Sensitivity

- The heatmap maps HSR competitiveness across two policy levers: carbon price (€50–300/t) and airport boarding overhead (0.75–2.0 h)
  - **Baseline** (€100/t, 0.75 h overhead): flight is still preferred - HSR loses ~X min vs flying
  - **Tipping point on carbon:** at current EU ETS (~€130/t), HSR becomes competitive once overhead reaches ~1.25 h
  - **Tipping point on overhead:** Even at €100/t, a modest increase in airport friction to ~1.0 h flips the route to HSR-optimal
- **Real-world validation:** after the AVE high-speed line opened in 2008, BCN–MAD saw a ~60% reduction in air passengers (Eurostat), consistent with the model's prediction that the route si


Flight vs HSR: Carbon-Adjusted Cost by Distance
Carbon = €100/t | Overhead = 0.75 h | HSR = 250 km/h avg
Carbon-adjusted travel cost (hours)
Route distance (km)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
1400
Short-haul flight (time + carbon cost)
High-speed rail (time + carbon cost)
HSR advantage
Flight advantage
Break-even
~385 km
BCN–MAD
483 km
Data: GL2017 EU Rail, OpenFlights, Eurostat avia_paoa

Conclusions

- Only **187** of 10,694 short-haul routes have HSR access at both endpoints.
- At €100/t carbon price, HSR wins on cost up to ~**385 km** - covering **34** of those 187 routes.
  - Only 34 routes clear the threshold partly due to conservative model assumptions: the 0.75h overhead **understates** real airport friction, and the 5 km buffer likely **excludes** airports with indirect but viable HSR connections - suggesting the true substitutable set is larger
- BCN–MAD sits just above break-even: HSR dominates once overhead hits ~1.0 h or carbon price exceeds ~€130/t.

Limitations

- The 5 km buffer identifies geometric proximity to HSR, not actual station access - some airports may be miscounted as connected
- Carbon cost function uses a fixed value of time and a normative carbon price rather than market-calibrated parameters
- Travel elasticity θ is borrowed from trade literature, not estimated from transport data, so results are sensitive to this choice
- Rail network data dates from 2017 and may not reflect lines opened or closed since
- Counterfactual assumes full modal substitution on eligible routes; real-world shift is partial and gradual

Next Steps

- Replace geometric buffer with station-level data to confirm true airport-rail connectivity
- Calibrate θ from observed modal share data, using BCN–MAD as a natural experiment
- Extend passenger data beyond latest Eurostat year to capture post-COVID traffic recovery
- Build a route-level cost-benefit analysis incorporating ticket prices, frequency, and door-to-door journey time
- Develop a dynamic model that allows for induced demand and network frequency responses